

Dr. Chong Tang

Assistant Professor (Lecturer) in AI, University of Bristol | tangchong22@gmail.com

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Research Interests

Postdoctoral Research Fellow (PhD, University College London) working in **efficient and applied AI** for real-world multimodal and time-series data. I develop **robust, resource-aware deep learning** that remains reliable under noise, distribution shift, and deployment constraints, combining signal processing with foundation models and vision-language models. I have published **30+ peer-reviewed papers** (650+ citations) including at **NeurIPS** and **ICLR**. Alongside my research, I build **reproducible ML systems** (Docker-based environments, AWS experimentation, distributed training, and end-to-end data pipelines), and I contribute to teaching through guest lecturing and practical lab support. I am keen to join a research-focused School where I can **deliver high-quality core Computing and cloud/ML-systems teaching**, grow an **externally funded research programme**, and collaborate across Artificial Intelligence and Data Analytics and systems-oriented colleagues on scalable, trustworthy AI.

Education

University College London , PhD in Deep Learning-Assisted Perception, UK <i>Advisors: Prof. Kevin Chetty and Prof. Simon Julier</i>	Oct. 2019 – Aug. 2023
University College London , MRes in Robotics, UK	Sept. 2018 – Oct. 2019
University of Nottingham , BEng in EEE, UK	Sept. 2014 – Jul. 2018

Experience

Assistant Professor (Lecturer) in Artificial Intelligence <i>University of Bristol, School of Engineering Mathematics and Technology, UK</i>	2026 – Present Bristol, UK
Postdoctoral Research Fellow (Visiting) <i>University College London, UK</i>	Sept. 2024 – Present London, UK
• Supervisor: Dr. Jagmohan Chauhan. • Developed compute-efficient training and inference techniques for foundation models on multimodal/time-series data, targeting robustness under noise, distribution shift, and practical deployment limits. • Implemented and evaluated token-efficient and sparse inference schemes for VLM/LLM-based video understanding, enabling training-free acceleration with systematic ablations and benchmarking. • Maintained a reproducible large-model experimentation stack (containerised environments, automated run configurations, and distributed execution when needed) to support reliable results and rapid iteration.	
Postdoctoral Research Fellow <i>University of Southampton, UK</i>	May 2023 – Sept. 2024 UK
• Supervisor: Dr. Jagmohan Chauhan. • Developed continual and multi-task learning methods for streaming sensor signals, improving robustness to distribution shift while reducing compute/memory requirements for long-running deployments. • Led implementation and optimisation of transformer-based pipelines, including profiling-driven performance tuning and memory-efficiency improvements for deployment on resource-constrained platforms. • Created end-to-end data pipelines for model training and evaluation (data ingestion, preprocessing, experiment tracking, and reproducible runs), enabling reliable comparison across datasets and settings.	

Selected Publications

- [1] **Chong Tang**, et al., SURGE: Surprise-Guided Token Reduction for Efficient Video Understanding with VLMs, **ICLR 2026**
- [2] **Chong Tang**, et al., PhySwin: An Efficient and Physically-Grounded Foundation Model for Multispectral Earth Observation, **NeurIPS 2025**
- [3] **Chong Tang**, et al., AdaTM: logic inspired adaptive Tsetlin machines for efficient and effective continual learning on the edge, **EWSN 2024**
- [4] **Chong Tang**, et al., MDPose: Human skeletal motion reconstruction using WiFi micro-Doppler signatures, **IEEE Transactions on Aerospace and Electronic System 2023**
- [5] **Chong Tang**, et al., FMNet: Latent feature-wise mapping network for cleaning up noisy micro-Doppler spectrogram, **IEEE Transactions on Geoscience and Remote Sensing 2022**

Projects

Years	Project & Sponsor	Budget	Contribution
2024–26	Perfect Recollection for Clearer Insight (EP-SRC)	£9.62 M	Postdoctoral Researcher ; Efficient foundation model and VLM design
2023–25	Multi-Tasking & Continual Learning for Audio Sensing on Resource-Constrained Platforms (EPSRC)	£4.23 M	Postdoctoral Researcher ; Lightweight continual-learning algorithms
2021–26	COG-MHEAR: Cognitively-Inspired 5G-IoT Multimodal Hearing Aids (EPSRC)	£30.23 M	Research Assistant ; RF-guided multimodal fusion framework
2019–23	OPERA: Opportunistic Passive Radar for Contextual Sensing (EPSRC)	£12.49 M	Research Assistant ; AI-driven RF sensing algorithms
2023–24	Radar-Based Situational Awareness for WIN ATMs (UCL Consultants)	£7.30 K	Collaborator ; Embedded ML deployment & optimisation

Teaching Experience & Supervision

- **Guest Lecturer**, University of Southampton, Department of ECS
 - Delivered invited MSc lectures on Internet of Things, with demonstrations and Q&A/tutorial support.
- **Teaching Assistant**, University College London (2020–2022)
 - COMP0130 Robot Vision and Navigation (PG): ran experiments and coding sessions, provided formative feedback, and supported assessment/marketing.
- **Project Co-supervision** (2020–2024)
 - Co-supervised MSc and PhD projects at UCL, University of Glasgow, and University of Southampton (deep learning, efficient AI, multimodal/time-series modelling).
- **Teaching coverage**
 - AI/Data: Machine Learning, Deep Learning, Applied AI/Data Analytics.
 - Core & cloud/systems: core computing foundations and cloud/systems topics including containerised workflows, distributed training, and scalable ML pipelines on cloud/HPC; with performance- and memory-aware optimisation experience relevant to systems/PL-oriented teaching.

Professional Service

- **Program Committee Member**
 - Program Committee Member, **ICMR 2026**
 - Shadow Program Committee Member, **EuroSys 2026** (European Conference on Computer Systems)
 - **MobiUK 2024** (Mobile, Wearable and Ubiquitous Systems, UK)

- **Journal Reviewer** for IEEE Trans. on Mobile Computing, IEEE Sensors Journal, IEEE Internet of Things Journal and IEEE Journal of Biomedical and Health Informatics.
- **Conference Reviewer:** NeurIPS, ICLR, ICML, ACM MultiMedia, IEEE PerCom, UbiComp, and related venues.
- **Professional Memberships:** Association for Computing Machinery (ACM)

Selected Awards

NVIDIA Academic Grants – Cloud-Edge Collaborative Data Generation Framework for VLA Learning, 2025.

UK Royal Academy of Engineering IC Postdoctoral Research Fellowship – Finalist, 2023.

IEEE Radar Conference – Best Popular Outcome Award, 2nd Prize Self-Built Radar Challenge, 3rd Prize AI-Assisted Radar Signal Processing, 2022.

IET Radar, Sonar & Navigation – Highest Annual Downloaded Paper, 2022.